MILITARY SPECIFICATION

SWAB, CULTURE, Calcium Alginate

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

* 1. SCOPE

1.1 Scope. This specification covers sterile, nonbacteriostatic, calcium alginate tipped culture swabs, suitable for obtaining cultures in military medical treatment centers.

1.2 Classification. Culture swabs covered by this specification shall be of the following types, as specified (see 6.2).

Type I - Aluminum shaft
Type II - Wood shaft

* 2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

MMM-A-250	Adhesive, Water Resistant (for Sealing Waterproofed Paper).
PPP-B-566	Boxes, Folding Paperboard.
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.
PPP-B-676	Boxes, Set-Up, Paperboard.
PPP-T-76	Tape, Pressure-Sensitive Adhesive Paper (for Carton Sealing).

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Military

MIL-L-10547	Liners, Case and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.
MIL-S-36586	Sterilization Test Strip Set, Bacterial Spore.

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

(Copies of Military specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

U.S. PHARMACOPEIAL CONVENTION, INC.

Pharmacopeia of the United States

(Application for copies should be addressed to the Mack Publishing Company, Easton, Pa. 18042.)

3. REQUIREMENTS

3.1 Material. Material shall be suitable for the purpose intended and as hereinafter specified.

3.1.1 Calcium alginate. Calcium alginate shall be soft, wool-like, non-toxic, pure medicinal grade fiber, and shall be soluble when tested as specified in 4.4.1

3.1.2 Aluminum. Aluminum shall be soft and flexible.

3.1.3 Wood. Wood shall be tough and flexible, and shall be gumwood, birch, or other wood suitable for the purpose intended. The wood shall be free of cross grain, knots, stains, and fungus growth.

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3.2 Construction.

3.2.1 Type I. Type I swab shall have a smooth aluminum wire shaft. One end of the shaft shall have a loop and the other end shall be covered with a calcium alginate tip.

3.2.2 Type II. Type II swab shall have a smooth round wood shaft, with a pear-shaped calcium alginate tip secured to one end. The tip shall cover the end of the stick and shall extend a minimum of 1/16 inch beyond the end of the stick.

3.3 Dimensions. Dimensions shall be in accordance with table I.

Table I. Dimensions

	Type I	Type II
Overall length	5-5/16 ± 1/8 inch	5-7/8 ± 1/8 inch
Shaft diameter	0.035 ± 0.003 inch	1/12 ± 1/32 - 1/64 inch
Tip length	9/16 nominal	
Tip diameter	1/8 ± 1/16 inch	

* 3.4 Sterility. The culture swabs shall be sterile when tested with a biological indicator as specified in 4.4.2.

3.4.1 Certificate of sterility. A certificate of sterility, stating that samples from each sterilizer lot have been tested and found to be sterile, shall be submitted to the procuring activity. The certificate of sterility shall include:

a. Item identification.
b. Federal stock number.
c. Contract or purchase order number.
d. Sterility lot or control number.
e. Date of sterilization.
f. Quantity of product in sterilizer lot.
g. Type of sterilization (gas, radiation, heat, steam, etc.).
h. Form of biological indicator (i.e., inoculated paper spore strip or carrier, inoculated product, or inoculated simulated product).
i. Test organism(s) and population density used.
j. Test medium, incubation temperature, incubation time, and dates of test.
k. Number of test units.
l. Statement that samples comply with sterility tests.
m. Signature of head of laboratory or a designated qualified alternate.

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When inspection is made at destination, the certificate of sterility shall be submitted by the contractor to the Directorate of Medical Materiel, ATTN: DPSC-ATQ, Defense Personnel Support Center, 2800 South 20th Street, Philadelphia, Pa., 19101, and a copy forwarded with each shipment to the consignee. When inspection is made at source, certificate of sterility shall be furnished to the cognizant Government Quality Assurance Representative for submission to the procuring activity.

3.4.2 Sterility lot. A sterility lot is that single quantity of product subjected to the same manufacturing operation and simultaneously sterilized in the same sterilizer chamber. Each unit package shall be clearly marked with a sterility lot (control) number and date of sterilization as specified in 5.3.1. Each sterility lot shall be traceable to the pertinent manufacturing lot(s).

3.5 Bacteriostatic activity. The calcium alginate fiber shall show no bacteriostatic activity, when tested as specified in 4.4.3.

3.6 Workmanship. Workmanship shall be first class throughout. Culture swabs shall be free of defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual, or auditory, investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

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4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and table II.

Table II. Sampling for examination

	Unit of product	Inspection level	AQL (percent defective)
For visual examination			
Major A defects	Envelope	II	1.0
Major B defects	Swab	II	2.5
Minor A defects	Unit carton	II	4.0
Minor B defects	Envelope	II	4.0
Minor C defects	Swab	II	4.0
For dimensional examination	Swab	S-2	2.5

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and table III.

Table III. Tests

Component (unit of product)	Characteristic	Requirement	Test procedure	Insp. level	AQL
Swab	Solubility	3.1.1	4.4.1	S-1	*
Envelope plus components	Sterility	3.4	4.4.2	S-1	*
Swab	Bacteriostatic activity	3.5	4.4.3	S-1	*

* Acceptance number shall be zero.

4.3 Examination. The swabs and envelopes shall be examined to determine compliance with all requirements contained in this specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with the following classification of defects.

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Table IV. Classification of defects

Categories	Defects*
Major A:	
101	Envelope not free of tears and holes.
102	Culture swabs not free of dirt or foreign matter.
Major B:	
151	Shaft not free of cracks and breaks.
152	Shaft not free of slivers.
153	Shaft not free of kinks (type II).
154	Tip not secured to shaft.
155	Tip not pear-shaped (type II).
156	Tip not free of fraying.
Minor A:	
201	Marking of unit carton not complete, correct, or legible.
Minor B:	
231	Marking of envelope not complete, correct, or legible.
Minor C:	
261	Surface of shaft not smooth.
262	Shaft not free of stains and knots (type II).

* Examination shall not be restricted to the defects listed above.

4.3.2 Dimensional examination. The culture swabs shall be examined for defects in dimensions. Any dimension not within the tolerances specified herein shall be classified as a defect.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Solubility. A separate sample of calcium alginate shall be placed in each of the following aqueous solutions:

a. 1 percent sodium citrate
b. 1 percent gylcerophosphate, and
c. 1 percent sodium hexametaphosphate in 1/4 strength Ringers solution (USP).

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* 4.4.2 Sterility test. The culture swabs shall be tested for sterility in accordance with the method specified in the U.S.P. for Purified Cotton, Gauze, Surgical Dressings and related material using inoculated product, inoculated simulated product, or paper spore strip (carrier). Product containing a biological indicator shall be marked or labeled so that it clearly differentiates from product intended for distribution. The incubation medium and minimum time in each case shall be soybean casein digest for 7 days for inoculated product, inoculated simulated product, or paper spore strip (carrier). The incubation temperature shall be as indicated in paragraph 4.4.2.1. The minimum number of test units, regardless of lot size, shall be 10, except for inoculated carrier which shall be 20 for Type I Sterilization and 30 for Type II Sterilization including inoculated carrier and product test units. The inspection level shall be as indicated in table III, Sampling Criteria for Tests.

4.4.2.1 Inoculated product or simulated product. For each type of sterilization, the organism used shall be as listed below:

Type of Sterilization	Organism	Incubation Temperature
Steam	B. stearothermophilus	55° - 65°C.
Dry Heat	B. stearothermophilus, or	55° - 65°C.
	B. subtilis var. niger (or globigii)	30° - 35°C.
Gas	B. subtilis var. niger (or globigii)	30° - 35°C.
Radiation	B. pumilus	30° - 35°C.

The number of viable cells shall be sufficient to provide assurance that the product has been sterilized.

4.4.2.2 Inoculated carrier. Inoculated carriers (paper spore strips) if used shall be in accordance with MIL-S-36586. Other paper carriers shall be inoculated to be of equivalent resistance. The method of incubation and number of test units shall be as indicated in 4.4.2 Sterility test and 4.4.2.1 Inoculated product or simulated product. The method of sterilization will control the organism in the spore strip to be tested for as shown in 4.4.2.1. NOTE: When an inoculated carrier is chosen, product samples must also be tested as required by the U.S.P.

4.4.2.3 Alternate method. Suppliers conducting sterility tests or designating lots in a manner other than that indicated herein shall submit proof that the alternate method provides equal assurance concerning the sterility of the product. However, all sterility tests shall be conducted using biological indicators. Requests for approval of any alternate method of testing for sterility shall be made to the Contracting Officer, and approval obtained before employment of the alternate method on a government contract.

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4.4.2.4 Product testing. Testing for sterility of the product after receipt by the government shall be without a biological indicator and in accordance with the U.S.P.

4.4.3 Bacteriostatic activity. The calcium alginate shall be tested for bacteriostatic activity, as follows:

a. 15 to 20 cubic centimeters (cc.) of nutrient agar shall be melted, poured into a dish, and cooled to 43.5 degrees Centigrade (° C.) ± 1.5° C.
b. Agar shall be inoculated with 0.1 cc. of a 24-hour nutrient broth culture* of Staphylococcus aureus, with the broth spread throughout the surface of the agar.
c. Three tips shall be equidistantly placed in intimate contact with the surface of the agar.
d. Agar plates shall be incubated for 24 to 48 hours under unglazed porcelain tops at 37° C.

*Test microorganism, Staphylococcus aureus, shall be propagated on nutrient agar and transferred once a month. Test microorganism shall be transplanted for not less than two days before inoculating in nutrient broth for test purposes.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be Levels A or C as specified (6.2).

5.1.1 Unit of issue. One box containing one hundred type I swabs, or fifty type II swabs, as specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in table V shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table V. Packaging quantities

Types	Unit package	Intermediate package	Exterior container
Type I	1 unit	6 units	72 units
Type II	1 unit	12 units	144 units

5.1.2.1 Packing variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

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5.1.3 Level A.

5.1.3.1 Unit package.

5.1.3.1.1 Envelope. One type I swab or two type II swabs shall be packaged in a sealed, peel-type envelope of appropriate size and design. Each envelope shall be capable of maintaining sterility of contents unless damaged or opened.

5.1.3.1.2 Unit carton. Twenty-five type II or one hundred type I envelopes as specified in 5.1.3.1.1 shall be packaged in a box of appropriate size and design conforming to PPP-B-566 or PPP-B-676, except that commercial colors will be acceptable. Closure shall be adequate to prevent spilling of contents under normal handling.

5.1.3.2 Intermediate package. Intermediate package shall be a box of appropriate size and design constructed in accordance with PPP-B-636, type CF, class weather-resistant. Boxes shall be closed by taping the top and bottom flaps with 2-inch wide tape conforming to PPP-T-76. Closing of bottom flaps with adhesive conforming to MMM-A-250 or metal stitching is permitted.

5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

* 5.2 Packing. Packing shall be Levels A, B or C as specified (see 6.2).

5.2.1 Level A.

* 5.2.1.1 Exterior container. Exterior container shall be designed for type 2 load, and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-601, class 2; or PPP-B-636, grade V3c. Closure and strapping shall be in accordance with appendix of applicable box specification. Special requirements of applicable table shall apply for PPP-B-636 box.

5.2.1.2 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes. Each fiberboard box shall be waterproofed in accordance with 30.4 of PPP-B-636.

5.2.2 Level B.

* 5.2.2.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, type CF, class domestic. Closure of wood boxes shall be as specified in the appendix of the applicable box specification. Closure of fiberboard boxes shall conform to method II as specified in PPP-B-636. Special requirements of applicable table shall apply for PPP-B-636 box.

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5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Envelope. In addition to any commercial markings, each envelope shall be marked to identify contents. In addition, marking shall include name of contractor, lot (control) number and the legend: "STERILITY GUARANTEED UNLESS ENVELOPE IS DAMAGED OR OPENED".

5.3.2 Unit package. Each unit package shall be marked in accordance with MIL-STD-129. Lot (control) number, contract or purchase order number and name of contractor shall be shown. In addition, marking shall indicate that the contents are non-toxic, sterile, nonbacteriostatic , and also the chemical solutions in which swab tips are soluble. The method of sterilization (G-Gas, R-Radiation, S-Steam, D-Dry heat sterilization) shall be shown following the level and date of packaging (Example: "A-6/71-G"). Final sterilization shall be on or after indicated packaging date.

5.3.3 Intermediate package. Each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Lot (control) number, contract or purchase order number and name of contractor shall be shown. In addition, the method of sterilization as specified in 5.3.2 shall be shown.

5.3.4 Exterior containers. Exterior containers shall be marked in accordance with MIL-STD-129. Lot (control) number shall be shown. In addition, the method of sterilization as specified in 5.3.2 shall be shown.

6. NOTES

6.1 Intended use. The culture swabs are intended for use in obtaining cultures from the nasopharynx.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Federal Stock Number (FSN).
c. Type required (see 1.2).
d. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

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6.3 This specification does not cover all types, classes, grades, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types which are normally procured to meet requirements.

6.4 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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MIL-S-36860

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CONTRACT NUMBER

MATERIAL PROCURED UNDER A
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1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
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